

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with 6+ years building **Node.js** and **TypeScript** APIs on **Google Cloud** (Firestore, Pub/Sub, Cloud Functions, GCS), with working AWS experience. Led the zero-downtime migration of **20+ production applications** to Firebase/GCP, including a real-time two-way MongoDB↔Firestore sync where **correctness was non-negotiable**: idempotent, redelivery-safe writes kept both stores consistent while both served live traffic. Repeatedly took over **services I did not write** and made them cheaper and faster (cut \$5,000/month of cloud spend; optimised a pricing backend over 2M+ records), then **automated and documented** the work so Ops and other engineers could run it without me. Design external integration surfaces (gRPC) and run an **agentic AI pipeline** daily with explicit boundaries on what it may do unattended.

CORE COMPETENCIES

Node.js & TypeScript APIs

GCP: Firestore, Pub/Sub, Cloud Functions

Data Consistency & Idempotent Sync

Zero-Downtime Migrations

Service Hardening & Cost Reduction

B2B API Integrations (gRPC)

Docker & Kubernetes

AI-Agent Workflows

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote · reporting to the CTO

- Built a **real-time two-way sync** between MongoDB and Firestore in **Node.js** on Google **Pub/Sub**, keeping two live datastores consistent under production writes from both sides — idempotent handlers, redelivery-safe writes, no duplicate or lost records.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/**GCP** with **zero downtime**, cutting services over one at a time and verifying each against live database state before retiring the old path.
- Took all remaining services off **AWS** (S3/EC2/EBS) onto GCP, scripting the S3→GCS storage move in **Python**; cut infrastructure spend by **\$5,000/month** and removed a second cloud from the operating surface.
- Shipped revenue-bearing backend features: Express Shipping (**+\$40 revenue per order**) and a 3D visualisation feature (video overlay, ffmpeg) that increased buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Inherited a Parse/MongoDB backend I had not written and built the **Node.js** migration pipeline that read, transformed and wrote records into Firestore with complex processing logic.
- Gave the platform away**: wrote the Firebase SDK documentation (Node.js and Python) for the other engineers and trained the Ops team on the Firebase dashboard and backup procedures so the system no longer needed me to run it.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Extended backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js for a real-time stock-price PWA.
- Optimised an inherited Firebase backend: built jobs querying **2M+ Firestore records** to keep customer-facing prices current from Morningstar APIs; added responsive React/Tailwind pages.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built **Node.js** backends for DeFi applications (Web3) and delivered React UIs from Figma under tight agency timelines, reporting to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected a microservices backend for FIDA Uganda's case-management system on **Apache Kafka, Docker and Kubernetes**; led a team of 4 developers.
- Designed a **gRPC integration surface** that multiple external USSD legal-aid providers built against a single **Node.js** backend; delivered a paralegal database (LASPNET) and a React Native/Expo mobile app with Firebase push notifications.

PROJECTS

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore sync with idempotent, redelivery-safe writes and origin tagging to prevent echo loops; ran under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · Firestore · MongoDB

career-ops — Agentic Evaluation Pipeline AI agents

Daily-use agent pipeline on Claude Code that scans job boards, verifies postings are live, extracts and scores them, and drafts tailored CVs. Unattended: scanning, dedup, verification. Reviewed: every score and document. Never autonomous: submission — a human owns Send.

Node.js · Playwright · Claude Code · YAML

Healthcare WhatsApp Chatbot

Conversational assistant integrating OpenAI and Google Vision for prescription OCR over the WhatsApp Business API.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

gRPC Service Bus for USSD Providers

Schema-first, versioned gRPC contract letting multiple third-party USSD providers integrate against one legal-aid backend with no per-provider code.

gRPC · Protobuf · Node.js · Apache Kafka

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Backend: Node.js, TypeScript, Express.js, REST & gRPC APIs, Microservices, Firebase Cloud Functions

Cloud: GCP (Firestore, Pub/Sub, Cloud Functions, GCS, Firebase), AWS (S3, EC2, EBS — working knowledge)

Infra: Docker, Kubernetes, Apache Kafka, NATS Streaming, Linux, Git **Data:** MongoDB, Firestore, SQL

Languages: JavaScript, TypeScript, Python, Go (intermediate), Rust (intermediate) **Spoken:** English, Kiswahili